A Simulated V1 population and decoding pipeline

Encode

Von Mises–tuned units, Poisson spike counts, contrast-dependent response gain (γ_c) only — contrast does not stretch each unit's tuning curve.

→

Decode choice / accuracy

Fixed linear decoder, $e = w^\top z$ (tuning-slope weights). Choice = $1\{e>0\}$; accuracy from sign(e).

→

Read out confidence

Six computations, each a function of e alone or of the full response r (below); accuracy itself is a seventh, model-free benchmark.

Each session simulates a large (N=500) population that generates choice, accuracy, and all six confidence read-outs; only a randomly drawn, per-session-realistic-sized *recorded* subsample of it is analyzed by the chosen/unchosen pool-weighting fit below, mirroring how a real recording samples only a small fraction of the animal's decision-relevant population. All seven targets are then scored the same way: the chosen/unchosen pool-weighting fit (Methods), summarized by −b/a (Figure 5). This is a fixed, illustrative simulation — nothing here is fit to the animals' behavior.

B Three routes to a confidence read-out, one simulated population

Route 1 — scalar e + reliability

Normalize |e| by a (known, estimated, or pooled) noise scale.

↔ Scalar-e family, Fig. S3

Bayesian(|e|)

$p = \Phi(|e|/\sigma_c)$

σ_c known exactly — the posterior probability correct given e and contrast.

ContrastBlind

$p = \Phi(|e|/\sigma_{pool})$

Same computation, but reliability is not conditioned on contrast — a simpler heuristic.

CASANDRE

$p = \Phi(|e|/\hat{\sigma})$, $\hat{\sigma} = \sigma_c \cdot e^{\eta\varepsilon}$

Reliability itself is noisily estimated each trial (log-normal meta-noise, η=0.7).

Route 2 — full posterior from r

Condition on the entire simulated response, not only its projection e.

↔ Posterior family, Fig. S3

FullBayes(r)

$P(s_k, c_j \mid r) \propto \prod_i \text{Poisson}(n_i; \lambda_i)$

Exact posterior over the stimulus×contrast grid; reports the sign-matching mass.

Entropy

$-\sum_{k,j} P(s_k, c_j|r) \log P(s_k, c_j|r)$

A distributional summary of the same posterior's overall sharpness, not its sign-specific mass.

Both use the identical posterior; Entropy predicts accuracy distinctly worse (AUC 0.925 vs. 0.977) despite sharing it with FullBayes(r) — but this weaker link to accuracy does not, on its own, pull Entropy toward the chosen-pool-only pattern (see Result below).

Route 3 — chosen/unchosen pools

Build confidence directly from the two pools, not from e or the posterior.

↔ Evidence-heuristic family, Fig. S3

PartialPositiveEvidence illustrative, α=0.5 fixed

$\text{conf} = E^{+}_{sim} - \alpha \cdot E^{-}_{sim}$

Evaluated here at a single α=0.5 (unchosen pool at half weight) — not fit to any data. α=0: pure positive-evidence bias; α=1: recovers (approximately) e itself.

Accuracy (model-free)

The trial's realized correct/incorrect outcome — a seventh benchmark, not a confidence computation.

Result (Figure 5): six of the seven targets, including Entropy, sit near the symmetric decision-variable line; only PartialPositiveEvidence approaches the chosen-pool-only pattern seen in the real neural confidence read-out, by explicit construction. Whether this discount read-out, fit rather than fixed, actually describes the animals' *behavior* is answered by the corresponding behavioral-fit family in Fig. S3 (Figures 6–7).

Reading this alongside Fig. S3

This figure asks whether a *fixed*, sensible read-out applied to a simulated population reproduces the chosen-pool-dominated pattern of the real neural confidence data — nothing here is fit to behavior. Fig. S3 asks the complementary question directly of the animals' actual confidence reports: which computation, allowed to vary its own parameters, fits best? The two agree qualitatively (an explicit chosen/unchosen asymmetry is needed) but only Fig. S3's fits (Figures 6–7) speak to the animals' behavior quantitatively.

Reading the figure

Φ: standard normal CDF. e: 1-D decision variable (real data) or its simulated analog $w^\top z$. E^{+},E^{-}: chosen/unchosen pool activity. σ_c, σ_{pool}: contrast-conditional and pooled reliability.

Describes the fixed, illustrative population simulation of Figure 5 (Methods, “Simulating a V1 population”). The behavioral fit set is Fig. S3 (Methods, “Free-parameter models of real confidence reports”).

Source: manuscript Methods (“Simulating a V1 population”) and Supplemental Information (identifiability discussion).